Ecotrons," he said. But first of all, what are Ecotrons, you may wonder?

BUILDING ECOTRONS AND ECOFROSTS WITH AI

In a nutshell, Ecozen's technologies are modernising the irrigation and cold chain sectors of Indian agriculture through climate-smart deep tech solutions. Their Smart AI and IoT-enabled solar pumping solution is called Ecotron, whereas their equally Smart AI and IoT-enabled solar-powered cold room for preventing agricultural waste is called Ecofrost. Both are portable, solar-powered tech solutions, and this is how they're disrupting the agricultural sector in India over the past decade, according to Gupta.

Ecotron smart solar pump controllers provide embedded AI, powerful motor controls, and enhanced 4G connectivity, Gupta highlighted. "With over 70,000 units installed across India, irrigating over 3 lakh acres of land, Ecotron is India's top solar pumping solution," he claimed.

If smart irrigation is increasing agricultural productivity, then Ecozen's Ecofrost solution is aimed towards managing the harvested produce for longer shelf life and reduced wastage. "Ecofrost helps maximise agricultural profitability by increasing shelf life, reducing waste, and expanding market access through proprietary tech for maintaining precise temperature and humidity levels for pre-cooling and storage applications. Ecofrost enables our customers to sell when prices are right and in geographies where favorable market conditions exist," Gupta added.

Don't forget all of this is powered through solar energy, including a thermal energy storage solution called Ice Core which provides Ecozen's customers with up to 30 hours of batteryless backup. InverterTech, another patent-pending technology, ensures

USING AI, OTHER CUTTING EDGE TECH

Ecozen uses AI with edge analytics to help keep systems running smoothly by diagnosing problems and running fixes without any server-side or manpower intervention. For example, Ecotrons constantly send electrical and mechanical data online for further processing, which helps them use physics and simulation models of motors and pumps to predict issues in advance. IoT devices, for example, enable sending alerts to users on system performance and maintenance requirements, like when the solar panels need to be cleaned. All this smart sensing technology, coupled with 4G connectivity, enables farmers to monitor and control Ecotron and Ecofrosts with user-friendly mobile apps. "Using data science and machine learning frameworks, all data being sent in by our devices are captured in our EcozenAI dashboard and served to customers in an easy-to-understand manner that enables analysis and aids decision making, right from device management to service requests and tracking," claimed Devendra Gupta.

stable cooling performance even during low sunlight conditions.

Bringing Ecotron and Ecofrost solutions together is EcozenAI, an internally built platform, Gupta explained. "We have loaded our solutions with intelligent sensing technologies, smart control, and communication capabilities, all powered by EcozenAI.

Using our AI, IoT, and data science components, it performs predictive diagnostics, preventative maintenance, and device management remotely. EcozenAI is also the foundation of our service platform," he summed up.

"Both of our products are powered by clean solar energy and have till date generated over 1 billion kWh of clean energy," according to Gupta, with presence in 20 states in India and 10 countries worldwide.

ECOZEN'S SUSTAINABLE IMPACT

This is the part that gave Devendra Gupta a lot of joy to discuss. It's what brings home the entire value proposition of his agri-tech startup in no uncertain terms, where sustainability is the name of the game.

Since the past 12 years of Ecozen's operations, Gupta told me, they've positively impacted over 1 lakh farmers in India. That's not all, they've also abated over "1 million tons of greenhouse gas emissions, reduced over 20,000 metric tons of food waste, and reduced over 33 crore litres of diesel consumption," all the time while irrigating over 3 lakh acres of land in the country's farming belt. Now that's commendable for any enterprise, let alone a technology one!

Proliferation of climate-smart technology gives a lot of satisfaction to Ecozen's team, according to CEO Devendra Gupta. "For example, recently in Maharashtra, we received orders of nearly 25 per cent from target markets while facing stiff competition from 8 other players. The adoption was mainly due to the superior operational and connected technology embedded into our Ecotron controllers," he claimed. "Farmers also reap the benefits of sustainability-driven tech solutions," Gupta said. "Through our partner SokoFresh in Kenya, an avocado farmer started storing his produce in Ecofrost to increase his avocados' shelf life," Gupta suggested, something the farmer couldn't do earlier, hence cutting down his agricultural waste in the process.

What began as an idea during their time spent at IIT-Kharagpur has now turned into an industry-leading smart-tech solution that's disrupting Indian agriculture and beyond, the trio at the helm of Ecozen is just getting started, Devendra concluded. d